

Undo the above change and build again to revert back to a sensible OOo install.

Pick up any other issue from qa.openoffice.org and hack on it.

Anyway, that was just an example—although, not at all useful. This article mainly focuses on implementing missing VBA properties and objects.

Adding VBA bits

As mentioned earlier, OOo is not fully interoperable with VBA, which is still under heavy development by Sun, Novell and others. So contributions are expected to improve interoperability.

`ooo-build/test/macro/missing_vbafeatures.ods` will give you an updated list of missing VBA features.

To add any VBA object or property, we need to modify two modules:

1. `oovbaapi` module—`idl` files, interface for objects
2. `sc` module—`hxx`, `cxx` files, actual implementation

Here is a small example on how a property is added to a VBA object.

Pick up any missing property from `missing_vbafeatures.ods`, say `Application.Sheets`. Implementation for the `Application.Worksheets` property is already available, so linking the `Sheets` property to the `Worksheets` property can fix the issue. Actually, `Sheet` differs from `Worksheet` in the sense that a sheet can be a worksheet or a chart. For the moment, let's consider both as the same.

Go to `oovbaapi/` directory, and add the following line to `ooo/vba/excel/XApplication.idl` just before the line that reads "any Worksheets([in] any aIndex)":

```
any Sheets( [in] any aIndex );
```

Now *build* and *deliver* the `oovbaapi` module. Next, go to the `sc` module, open the `source/ui/vbaapplication.hxx` file and add the prototype for `Sheets`:

```
virtual css::uno::Any SAL_CALL Sheets( const css::uno::Any& aIndex ) throw(
(css::uno::RuntimeException);
```

...which is similar to the `Worksheets` function.

Go to `source/ui/vbaapplication.cxx` and add the definition for the above function:

```
uno::Any SAL_CALL ScVbaApplication::Sheets( const uno::Any& aIndex )
throw( uno::RuntimeException )
{
    return getActiveWorkbook()->Worksheet(aIndex);
}
```

The `ooo-build/test/macro/` directory contains many `xls` and `ods` files to test VBA features. To test this hack, open any available Excel document and add the following VBA code in an existing or new module:

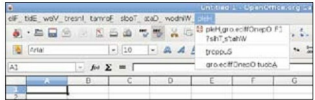


Figure 2: OOo hacked to make the names of the menu items appear in reverse order

```
MagBox Application.Sheets(1).Name
Application.Sheets(2).Name="Hello"
```

To add a new VBA Object, say `Sample`, create `XSample.idl` in `oovbaapi/ooo/vba/excel/`. Add the concerned entry in `makefile.mk`, then *build* and *deliver* in `oovbaapi`. Next, create `vbasample.hxx` and `vbasample.cxx` in `sc/source/ui/vba/` with an addition of `vbasample.o` to `makefile.mk`, followed by *build* and *deliver* in `sc`.

HelperAPI

HelperAPI is a parallel solution to support VBA in Star Office, which is Sun's proprietary offering based on OOo. But with the agreement of Sun and Novell to work together on VBA, Sun has donated HelperAPI and made it open now. HelperAPI is implemented in Java, whereas `oovbaapi` are C++ based. But still, understanding HelperAPI can help one to implement VBA better and avoid working from scratch.

So you now have the basic idea on adding the VBA property. I hope you can now start building Go-oo and add some other missing features. Patches can be sent to Noel Power (noel dot power at novell dot com), leader of the VBA Interoperability Project.

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